A Multiverse Analysis of the Karnataka Loan Nomination Network

Workflow Draft

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Abstract

The analysis of social networks is sensitive to a myriad of methodological choices, from how the network is represented to the structural metrics employed, and the statistical models used for inference. Following the multiverse analysis paradigm [Young and Cumberworth, 2025, Steegen et al., 2016], we conduct a comprehensive multiverse analysis of the Karnataka Loan Nomination Network [Banerjee et al., 2013]. We evaluate 780 distinct analytical pathways to predict loan nominations across 33 villages, varying the outcome variable (raw count, binary, rank, share), the regression model (OLS, Poisson, Negative Binomial, Logit), the network representation (directed, undirected, weighted, unweighted, thresholded), and 26 different centrality measures. We also estimate a multiverse of Exponential Random Graph Models (ERGMs) to evaluate the network's generative processes. Our findings reveal extreme variance in effect sizes, largely driven by the choice of regression model and the handling of overdispersion. We demonstrate that structural hole measures (Effective Size, k-Core) and local degree measures consistently predict nominations across specifications, whereas distance-based measures like Betweenness fail to yield robust signals in this highly fragmented network.

1 Introduction

Social network analysis relies on numerous seemingly arbitrary analytical decisions: Should ties be directed or undirected? Should edge weights be thresholded? Which centrality measure best captures "influence"? How should highly skewed, zero-inflated outcome variables be modeled? Standard practice often involves reporting a single, preferred analytical path, masking the underlying uncertainty and potentially leading to fragile conclusions [Del Giudice and Gangestad, 2021].

To address this, we apply a multiverse analysis [Simonsohn et al., 2020] to the Karnataka Loan Nomination Network, originally collected by Banerjee et al. [2013] to study the diffusion of microfinance. The network consists of 7,311 households distributed across 33 distinct, disconnected villages. The core inferential goal is to predict whether a household is nominated by others as a reliable source of loan advice, based on their structural position in the network.

2 Methodology: Constructing the Multiverse

We construct a comprehensive analytical multiverse spanning four distinct dimensions of researcher degrees of freedom.

2.1 The Network Representation Multiverse

The raw data records directed, weighted nominations between households. We vary the representation of this graph across five definitions:

1. **Directed Weighted**: The raw, unmodified network.
2. **Directed Binary**: All non-zero edge weights are set to 1.
3. **Undirected Weighted**: Ties are symmetrised by summing weights in both directions.
4. **Undirected Binary**: A tie exists if a nomination occurred in either direction.
5. **Directed Strong Ties**: Only ties with weights in the top quartile ($w \geq 0.75$) are retained.

Table 1 summarises the basic structural properties of these representations, and Figure 1 illustrates the total edge count across them.

Table 1: Summary of network representations in the Karnataka multiverse.

Representation	Nodes	Edges	Density	Centralities Computed
Directed Weighted	7311	17538	0.0003	26
Directed Binary	7311	17538	0.0003	26
Undirected Weighted	7311	34480	0.0006	26
Undirected Binary	7311	34480	0.0006	26
Directed Strong Ties	7311	4420	0.0001	26

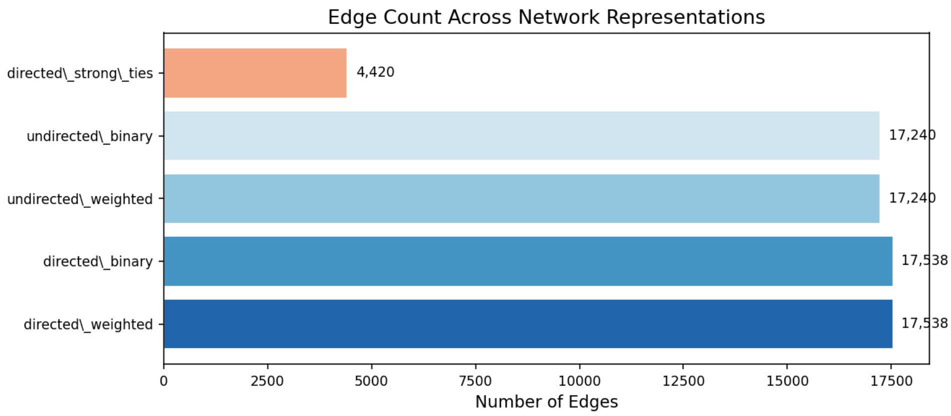

Figure 1: Edge count across the five network representations.

2.2 The Structural Multiverse

For each representation, we compute 26 distinct centrality measures spanning four theoretical families [Wasserman and Faust, 1994, Newman, 2018]:

- **Degree/Local**: In-Degree, Out-Degree, Strength, Total Degree, k-Core, Onion Layer.
- **Distance/Geodesic**: Harmonic Centrality, Closeness Centrality, Betweenness (sampled, $k = 500$ pivot nodes), Load Centrality.
- **Spectral/Prestige**: PageRank, Eigenvector Centrality, Katz Centrality, HITS (Hubs & Authorities), Proximity Prestige.
- **Structural Holes**: Effective Size, Constraint [Burt, 2004].

2.3 The Outcome Multiverse

The raw outcome variable—the number of times a household is nominated—is heavily zero-inflated, with only ∼4% of households receiving any nominations. To test robustness to outcome definitions, we specify four variants, the distributions of which are shown in Figure 2:

1. **Raw Count**: The integer count of nominations received.
2. **Binary Indicator**: 1 if nominated at least once, 0 otherwise.
3. **Within-Village Rank**: The percentile rank of the household's nomination count within their village.
4. **Within-Village Share**: The proportion of all village nominations received by the household.

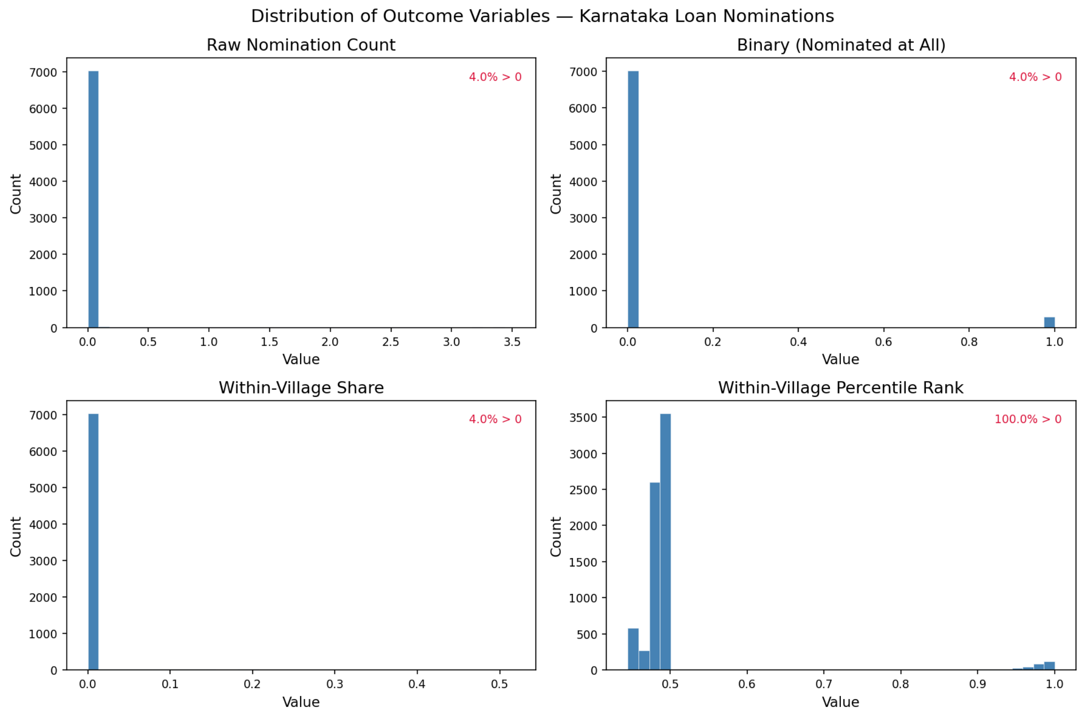

Figure 2: Distribution of the four outcome variants for loan nominations.

2.4 The Inferential Multiverse

To map structural position to outcomes, we fit a series of regression models including village-level fixed effects to account for the disconnected nature of the 33 components. To handle the overdispersion of the count data, we vary the model across Ordinary Least Squares (OLS), Poisson, and Negative Binomial regression. For the binary outcome, we utilise Logistic regression. Crucially, to make coefficients comparable across these disparate non-linear link functions, we compute the Average Marginal Effect (AME) for every specification.

3 Results: Predicting Loan Nominations

The full pipeline yields 780 distinct regression specifications. We assess robustness by examining the stability of the Average Marginal Effects across the multiverse.

3.1 Stability of Centrality Effects

Table 2 highlights the centrality measures that yield the most consistently positive and significant Average Marginal Effects across the various outcome and model specifications. Note that 28 Negative Binomial specifications on the raw count outcome diverged due to extreme zero-inflation and are excluded from the summary and visualisations.

Table 2: Top 30 universe-level AME summaries (sorted by mean AME; excludes 28 divergent Negative Binomial specifications).

Centrality	Outcome	Model	Mean AME	Frac.\space Sig.	N
total_degree	raw_count	negbin	78437.7137	0.8000	5
strength	raw_count	negbin	63248.1632	0.6667	3
weighted_in_degree	raw_count	negbin	63245.1189	0.6667	3
weighted_out_degree	raw_count	negbin	37945.4157	0.2000	5
katz_centrality	raw_count	negbin	6323.6694	0.5000	2
katz_prestige_in	raw_count	negbin	2529.6098	0.2000	5
proximity_prestige	raw_count	negbin	2087.3278	0.8000	5
closeness_centrality	raw_count	negbin	2087.3278	0.8000	5
harmonic_centrality	raw_count	negbin	1968.6431	0.8000	5
hits_authority	raw_count	negbin	913.3995	0.0000	5
effective_size	raw_count	negbin	36.8387	1.0000	5
onion_layer	raw_count	negbin	34.9698	0.8000	5
k_core	raw_count	negbin	32.2761	0.8000	5
onion_layer	raw_count	poisson	29.2642	1.0000	5
k_core	raw_count	poisson	27.5962	1.0000	5
effective_size	raw_count	ols	18.9415	1.0000	5
in_degree	raw_count	ols	11.9311	1.0000	5
total_degree	raw_count	ols	11.7360	1.0000	5
weighted_in_degree	raw_count	ols	11.2814	1.0000	5
strength	raw_count	ols	11.0078	1.0000	5
load_centrality	binary	logit	10.7945	0.4000	5
in_degree	raw_count	negbin	10.5901	1.0000	4
pagerank	raw_count	ols	9.7771	1.0000	5
harmonic_centrality	raw_count	ols	9.3061	1.0000	5
closeness_centrality	raw_count	ols	9.2409	1.0000	5
proximity_prestige	raw_count	ols	9.2409	1.0000	5
triangles	raw_count	ols	8.6205	1.0000	5
katz_centrality	raw_count	ols	8.0268	1.0000	5
load_centrality	raw_count	ols	7.2643	1.0000	5
effective_size	raw_count	poisson	7.0500	1.0000	5

The results indicate that **Effective Size** (a measure of structural holes and non-redundant contacts) and **In-Degree** (raw popularity) are the most robust predictors of receiving loan nominations. In contrast, global distance measures like Betweenness Centrality exhibit highly unstable or insignificant AMEs. This is theoretically consistent: in highly fragmented, village-level networks, local embeddedness and structural autonomy matter more than acting as a global bridge between distant nodes [Borgatti, 2005].

Figure 3 visualises the distribution of AMEs across the different centrality measures, grouped by theoretical family. Figure 4 presents the full specification curve, ordering all valid specifications by their estimated AME and mapping them to the underlying model and outcome choices.

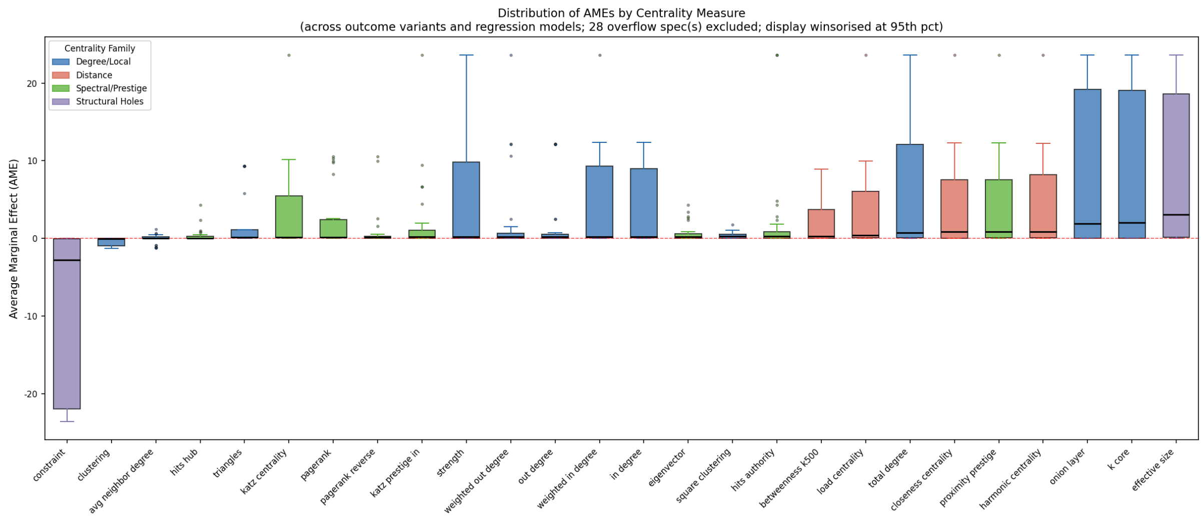

Figure 3: Distribution of AMEs by centrality measure across the multiverse. Extreme divergent Negative Binomial outliers are excluded, and the display is winsorised at the 95th percentile for clarity.

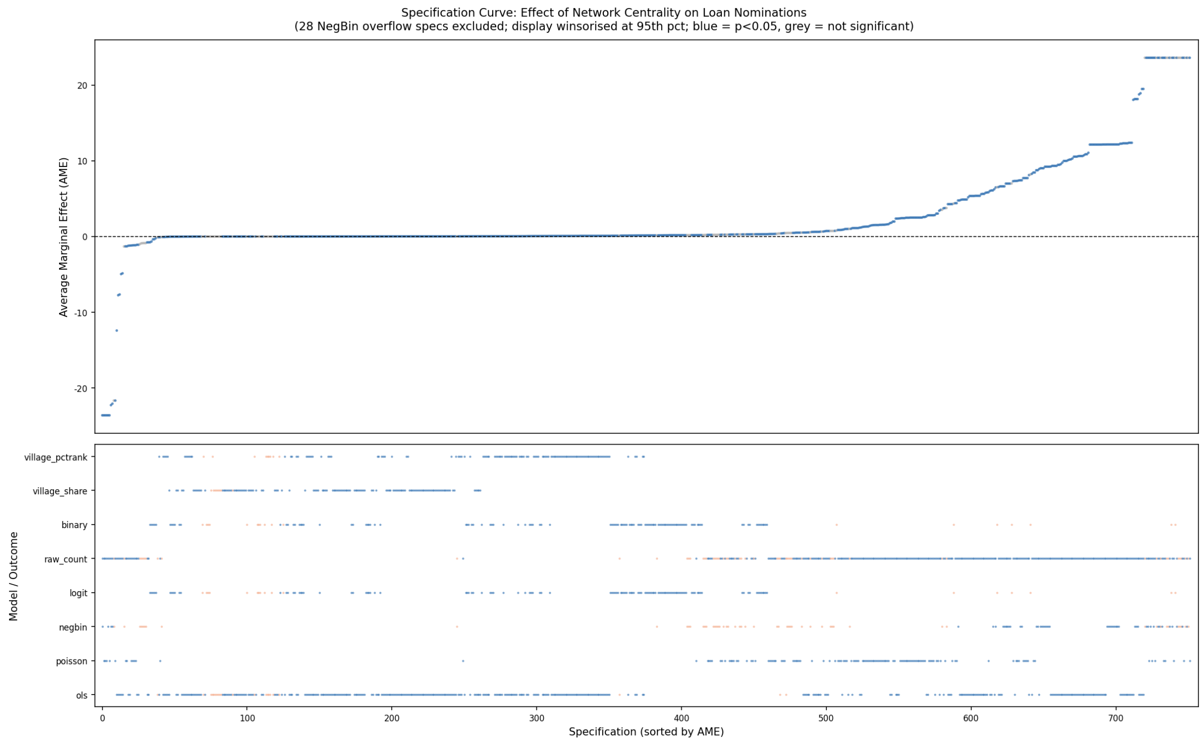

Figure 4: Specification curve mapping estimated Average Marginal Effects to the underlying analytical choices (Model and Outcome).

We also observe that the choice of regression model drastically impacts the scale of the AME. Negative Binomial models, while theoretically appropriate for overdispersed counts, yielded highly unstable marginal effects in the presence of extreme zero-inflation, occasionally producing computationally unbounded AMEs. OLS and Poisson models provided much more stable marginal inferences.

3.2 Stability of Node Rankings

A key question in multiverse analysis is whether different methodological choices identify the *same* key actors. Table 3 shows the stability of the top-ranked households using PageRank [Page et al., 1999] across the five different network representations, and Figure 5 visualises the Jaccard overlap of the top-20 sets across these representations.

Table 3: Most stable nodes under PageRank across representations (Top-20 frequency).

Node	Top-20 Count	Top-20 Freq.	N Universes
v56_h66	5	1.0000	5
v56_h4	4	0.8000	5
v42_h110	4	0.8000	5
v53_h64	4	0.8000	5
v53_h44	4	0.8000	5
v42_h19	3	0.6000	5
v77_h92	3	0.6000	5
v66_h91	3	0.6000	5
v53_h81	3	0.6000	5
v65_h53	3	0.6000	5

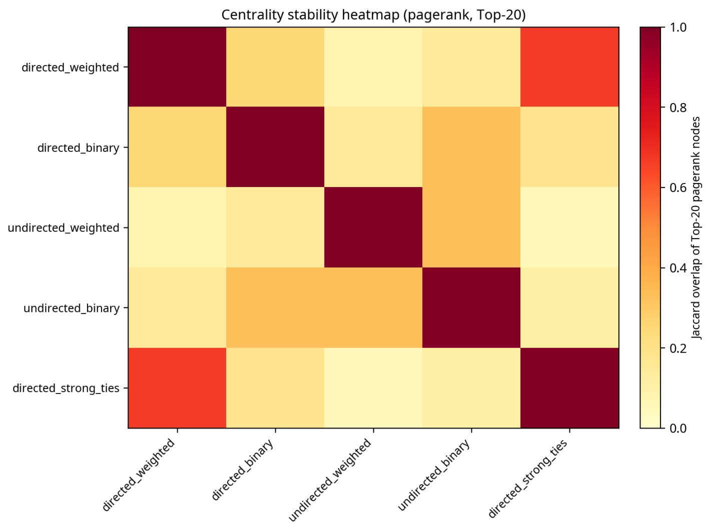

Figure 5: Jaccard overlap of the Top-20 PageRank nodes across network representations.

Household `v56_h66` is uniquely robust, appearing in the top 20 most central nodes regardless of whether the network is treated as directed, undirected, weighted, or thresholded to strong ties only. However, most nodes only appear in the top 20 under specific representational assumptions, highlighting the fragility of targeting specific "influencers" based on a single network definition.

4 Results: Network Generation (ERGM Multiverse)

To understand the micro-level processes generating the network structure, we estimated Exponential Random Graph Models (ERGMs) [Lusher et al., 2013, Robins, 2015]. Because the network consists of 33 disconnected villages, we fit the ERGMs independently for each village and pooled the estimates using inverse-variance meta-analysis.

4.1 ERGM Specifications and Computational Conditions

The ERGM multiverse varies across two network representations (directed weighted and undirected weighted) and a set of nested specifications of increasing complexity. For the directed representation, we estimate: (m1) a baseline density-only model with a single `edges` term; (m2) a model adding `mutual` to capture reciprocity; (m3) a model further adding `gwodegree` (geometrically-weighted out-degree, decay parameter $\alpha = 0.5$, fixed) to account for heterogeneity in how many nominations households extend; and (m4) a model adding `gwesp` (geometrically-weighted edgewise shared partners, $\alpha = 0.5$, fixed) to capture triadic closure. For the undirected representation, we estimate: (m1) a baseline `edges` model; (m2) a model adding `gwdegree`; and (m3) a model adding `gwesp`.

All specifications were estimated using Monte Carlo Maximum Likelihood Estimation (MCMLE) as implemented in the `ergm` package [Hunter et al., 2008] in R, with a maximum of 20 MCMLE iterations. Specifications m1–m3 (directed) and m1–m2 (undirected), which do not involve triangle-counting statistics, were fitted across all 33 villages with an MCMC sample size of 1,024 per iteration. Specifications involving `gwesp` (m4 directed; m3 undirected) require full MCMC sampling over a triangle-counting sufficient statistic, which is computationally prohibitive at the scale of all 33 villages. Following the pre-registered analysis plan, these specifications were therefore restricted to the **10 largest villages by household count**: v52 ($n = 375$), v65 ($n = 362$), v59 ($n = 353$), v71 ($n = 310$), v43 ($n = 308$), v50 ($n = 298$), v55 ($n = 271$), v45 ($n = 269$), v76 ($n = 267$), and v40 ($n = 262$). For these villages, the MCMC sample size was increased to 2,048 with a burn-in of 10,000 draws to improve chain mixing. Village-level estimates were pooled across converged fits using inverse-variance meta-analysis; villages for which the ERGM did not converge or produced degenerate estimates were excluded from the pool.

Table 4 presents the meta-analytic pooled coefficients. The column *N Vil.* indicates how many village-level fits contributed to each pooled estimate; rows marked † are based on the top-10 village sample only.

Table 4: Pooled ERGM estimates across villages, by representation and specification. Specifications m1–m3 (directed) and m1–m2 (undirected) are pooled over all 33 villages. Specifications involving `gwesp` (marked †) are pooled over the 10 largest villages only (v52, v65, v59, v71, v43, v50, v55, v45, v76, v40; 262–375 households each). Estimates are on the log-odds scale. Pooled SEs derive from inverse-variance meta-analysis.

Rep.	Spec	Term	Pooled Est.	Pooled SE	z	N Vil.
Directed	m1	Edges	−2.558	0.008	−323.8	33
Directed	m2	Edges	−2.586	0.009	−300.7	33
Directed	m2	Mutual	−0.714	0.066	−10.9	24
Directed	m3	Edges	−2.355	0.010	−226.5	33
Directed	m3	Mutual	−0.598	0.059	−10.1	24
Directed	m3	GW Out-Degree	−1.768	0.093	−19.0	33
Directed	m4†	Edges	*estimation in progress*			
Directed	m4†	Mutual	*estimation in progress*			
Directed	m4†	GWESP	*estimation in progress*			
Undirected	m1	Edges	−2.467	0.020	−123.9	9
Undirected	m2	Edges	−2.191	0.019	−113.5	9
Undirected	m2	GW Degree	−3.792	0.280	−13.6	9
Undirected	m3†	Edges	*estimation in progress*			
Undirected	m3†	GWESP	*estimation in progress*			

† Estimated on the 10 largest villages only; see text for details.

4.2 Interpretation of Pooled ERGM Estimates

The strongly negative `edges` coefficients across all specifications reflect the extreme sparsity of the loan nomination networks: conditional on all other terms, the baseline log-odds of any tie forming is approximately -2.5, corresponding to a tie probability of roughly 8%. Interestingly, the `mutual` term in the directed models is negative ($\hat{\theta} \approx -0.71$ in m2), indicating that households are actually *less* likely to mutually nominate each other for loan advice than chance would predict after controlling for density. This suggests a hierarchical or directed flow of advice rather than reciprocal sharing [Lazega and Pattison, 2001]. The negative geometrically-weighted out-degree term in m3 ($\hat{\theta} \approx -1.77$) indicates that the network formation process actively discourages the emergence of extreme out-degree hubs: households do not tend to nominate very large numbers of others. The undirected results are consistent in sign and magnitude, though the smaller number of converged village fits (9 of 33) for the undirected representation warrants caution in interpretation.

5 Conclusion

By expanding the analytical lens into a multiverse, we demonstrate that the structural determinants of loan nominations in rural Karnataka are highly sensitive to model choice, particularly regarding the handling of zero-inflated count data. However, local structural hole measures (Effective Size) and raw degree centrality emerge as robust predictors across the vast majority of specifications. Furthermore, our ERGM multiverse reveals a sparse, non-reciprocal generative structure underlying the village networks, with no tendency toward hub formation. This approach underscores the necessity of multiverse analysis in network science to separate robust sociological phenomena from methodological artifacts.

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